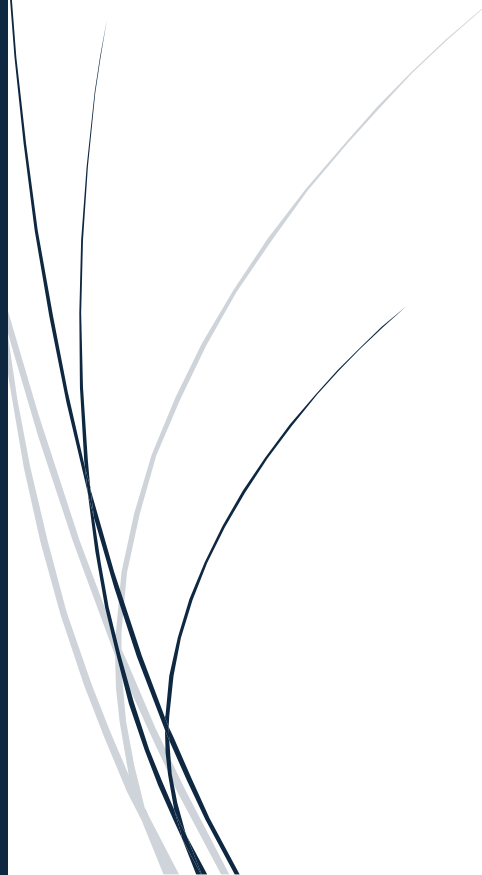




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Private Hospital Insurance in Australia: Long-Term Trends and Individual Determinants of Uptake



Paula Moreno

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Introduction

Australia runs a mixed health system, whereby the universal health care is provided through Medicare for the basic services offered by hospitals and doctors, whereas the PHI provides benefits such as reduced wait-times, choice of doctor, and improved facilities. This paper will analyze the trends of PHI enrollment and coverage in Australia to determine the extent to which the market was shrinking prior to the emergence of the pandemic and the dynamics that occurred in its operation after the outbreak. Data used in the report were obtained from the Quarterly Enrollment Report compiled by the APRA and population estimates from the Australian Bureau of Statistics covering the time frame of September 1997 and September 2025. The paper consists of two tasks, namely, task 1 which analyzes the trend and sustainability of the market and task 2 that formulates a regression model for the factors influencing an individual to enroll for PHI.

Task 1: Long-term trends in Private Hospital Insurance (PHI) uptake and the sustainability of the PHI market

The present study aims to investigate whether there has been any reduction in the number of PHI memberships and coverage ratios before the outbreak of the pandemic; see the different trends between age groups; analyze the behavior of the market during and post-pandemic; and determine the sustainability of the market.

Section 1: Data Exploration and Quality Assessment

This analysis draws on two primary data sources, the first is the "HT by Age" sheet from the APRA, PHI Membership Trends data, which records the number of individuals holding private hospital treatment insurance, broken down by five-year age group, for each quarter from September 1997 to September 2025. The second source is the ABS quarterly estimates of the Australian resident population by age group, covering the same time, these population figures were used alongside the APRA membership data to calculate the PHI coverage rate.

Figure 1: *PHI Coverage rate per quarter (first 10 rows)*

Final merged dataset shape: (2034, 5)
Time coverage: 1997-09-30 → 2025-09-30

Preview:

	Quarter	age_group	phi_members	population	coverage_rate
0	1997-09-30	0-4	317962	1292010	24.609871
1	1997-09-30	5-9	354654	1314746	26.975096
2	1997-09-30	10-14	390670	1305316	29.929151
3	1997-09-30	15-19	365913	1272745	28.749907
4	1997-09-30	20-24	233893	1342584	17.421107
5	1997-09-30	25-29	264023	1446757	18.249298
6	1997-09-30	30-34	341612	1416099	24.123455
7	1997-09-30	35-39	414953	1477936	28.076520
8	1997-09-30	40-44	444277	1376329	32.279855
9	1997-09-30	45-49	476474	1290924	36.909531

Figure 2: *Average coverage rate by age group (across all quarters)*

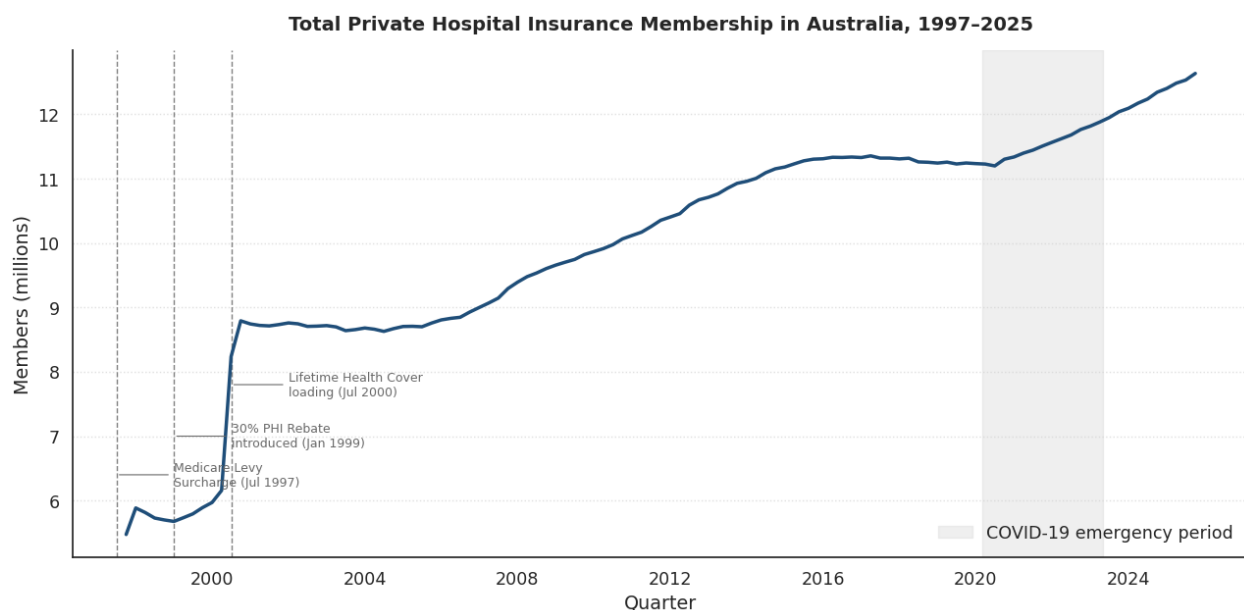
Average coverage rate by age group (across all quarters):	
age_group	
0-4	37.27
5-9	41.85
10-14	43.80
15-19	43.53
20-24	30.63
25-29	26.74
30-34	39.70
35-39	44.99
40-44	47.51
45-49	50.04
50-54	52.38
55-59	53.87
60-64	53.38
65-69	51.32
70-74	49.14
75-79	46.42
80-84	43.44
85+	39.10

The two data sets have been checked for any missing values, duplicates, or other discrepancies in the structure. Age groups have been standardized by putting together both data sets and categorizing them into five groups, that is: children (0-19), young adults (20-34), middle-age group (35-54), older but pre-retirement age group (55-69), and elderly group (70+). On the other hand, timelines were created for this analysis based on the information provided by WHO regarding their response to this pandemic (Timeline: WHO'S COVID-19 Response, n.d.). Pre-COVID (2015-2019), COVID (2020, mid 2023), post-COVID (mid 2023, 2025)

Section 2: Descriptive and Exploratory Analysis of PHI Trends

2.1 Long-Term Trend in Total PHI Membership: Has total PHI membership grown, declined, or stayed flat over time?

Figure 1: *Trend in total PHI membership (September 1997 to September 2025)*



According to Figure 1, total PHI membership rose steadily from about 5.5 million in September 1997 to 12.6 million in September 2025. The growth was uneven, however, with three sharp inflection points concentrated between 1997 and 2000, when three major policy reforms were introduced to lift PHI uptake (Palangkaraya & Yong, 2005). The Medicare Levy Surcharge (MLS), introduced on 1 July 1997, imposed an additional tax on higher-income earners without PHI. The 30% PHI Rebate, introduced on 1 January 1999, subsidized premium payments for all Australians. The Lifetime Health Cover (LHC) loading, introduced on 1 July 2000, penalized individuals who delayed purchasing PHI beyond age 30. Within three years, membership rose from approximately 6 million to 8.8 million a 45% increase.

From 2015 to 2019, years leading up to the COVID 19, memberships grew steadily but more slowly, reaching approximately 11.3 million for 2019. During the initial phase of the pandemic (2020 to mid-2021), membership growth temporarily stalled, however contrary to pre-pandemic

concerns about a potential market collapse, the enrollment accelerated significantly from 2022 onward, reaching 12.6 million by the end of the period 2025.

There are two important insights from the following graph. First, the government policies had a huge impact on PHI memberships, incentivising private cover reduces pressure and waiting times on the public system. Second This post-pandemic rebound reflects deferred demand, people who postponed purchasing PHI during the uncertainty of the pandemic re-enrolled once hospital services resumed.

2.2 Long-Term Trend in PHI Coverage Rate: How has the PHI coverage rate evolved once population growth is accounted for?

Figure 2: *Coverage Rate (September 1997 to September 2025)*

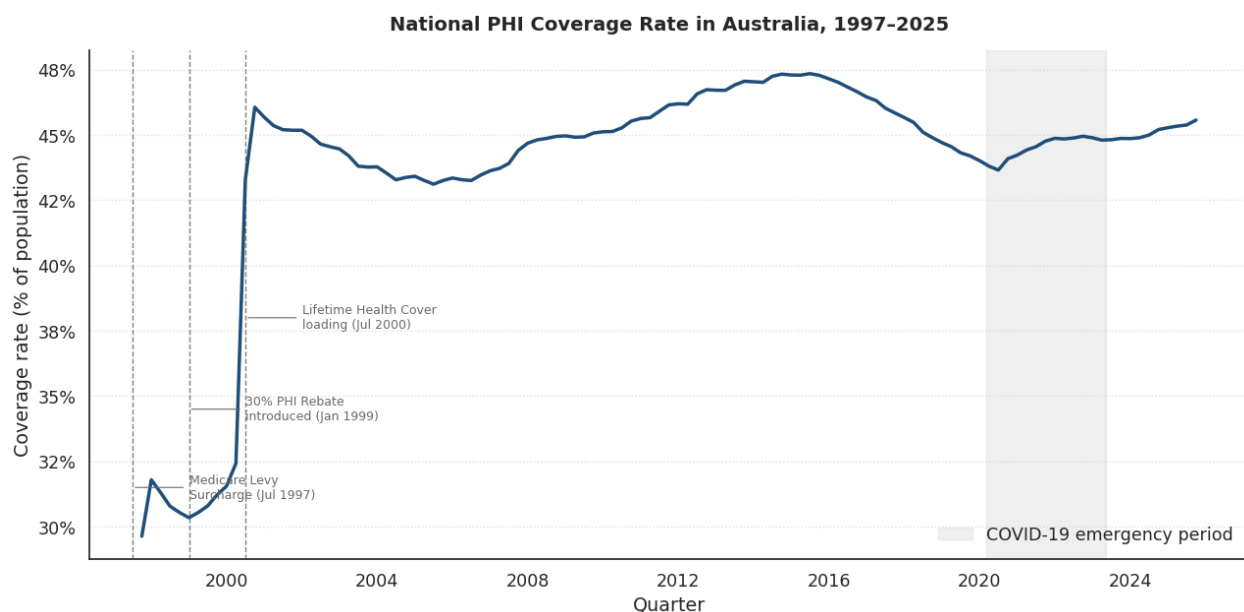


Figure 2 presents the national PHI coverage rate, the proportion of the Australian population holding private hospital treatment cover, which is a meaningful indicator of market health as it controls for population growth.

Between 1997 and 1999, the coverage rate was declining around 30%, but after the two government policy interventions of 1999–2000, the coverage rate reached a peak of 45.7%, a gain of nearly 15 percentage points in just three years, which shows us how impactful are government policies. From

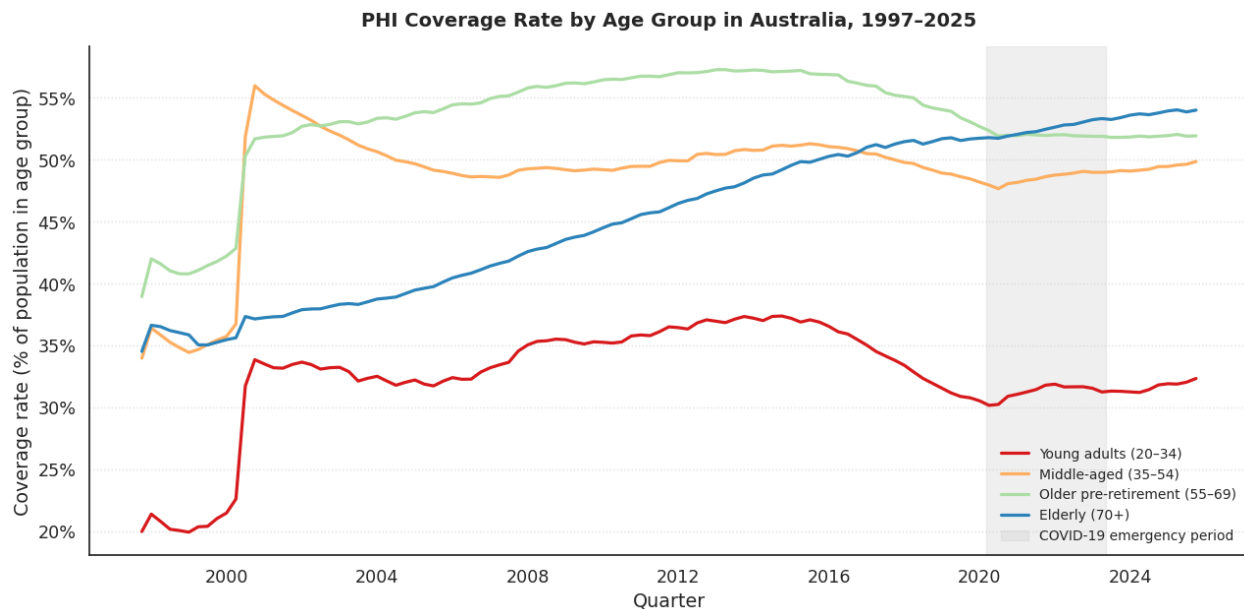
2001 onwards, coverage declined and rise along time, falling to 44.0% by December 2019, confirming that concerns about a weakening PHI market before the pandemic were true (Duckett & Moran, 2021). During 2020 and 2021, the rate fell further before recovering strongly from 2022 onwards, returning to 45.6% by September 2025. Table 1 summarizes these key movements.

Table 1: National PHI Coverage Rate at Key Points in Time (1999–2025)

Period	Coverage Rate
Pre-policy (Jun 1999)	30.79%
Post-policy peak (Dec 2000)	45.68%
Pre-COVID (Dec 2019)	44.02%
Latest (Sep 2025)	45.57%

2.3 Coverage Rate by Age Group: Did younger people drop out at a different rate from middle-aged and older people?

Figure 3: PHI Coverage Rate by Age Group (September 1997 to September 2025)



The 0–19 age group is excluded from this analysis, since PHI coverage for children and teenagers is typically determined by their parents, the analysis therefore focuses on the four adult age groups.

Figure 3 illustrates the disaggregation of the overall coverage rate for PHI by four different adult age. The most noticeable pattern is that observed in the young adult population aged 20-34 years; there was a significant improvement, their coverage rising from 20% to 33% after the policy

reforms implemented in 1999-2000, oscillating between 33% and 35% throughout the decade 2000s, reaching its peak around 37% in 2015, and subsequently dropping down to 30% by the end of 2019, before the pandemic outbreak.

The middle-aged (35–54) show long, gradual decay; their coverage has been slipping for over two decades, from 56% after the 2000 reforms to under 50% by 2025. There is no single turning point, just a long, steady drift away from private cover.

The older pre-retirement group (55–69) was the most stable, holding above 55% throughout the 2010s before a decline became visible after 2015, suggesting this was least sensitive to rising premiums given their anticipate need in hospital treatment and therefore more willing to pay rising premiums.

Only the elderly (70+) have shown an increase in coverage throughout the entire 28-year period, from about 35% in 1997 to more than 54% by 2025. In 2020, the graph of the elderly became higher than that of the older pre-retirement age group, meaning that currently, the group that is most insured is above the age of 70. The comparison of these two lines, reflects a significant change in the demographic structure of the insured population, with the aging of the insured population, the cost of claims becomes higher, leading to increased premiums.

2.4 Period Comparison: Pre-COVID, During COVID, and Post-COVID, How did the COVID-19 pandemic affect these trends?

Figure 4 : *Average PHI coverage rate across three periods*

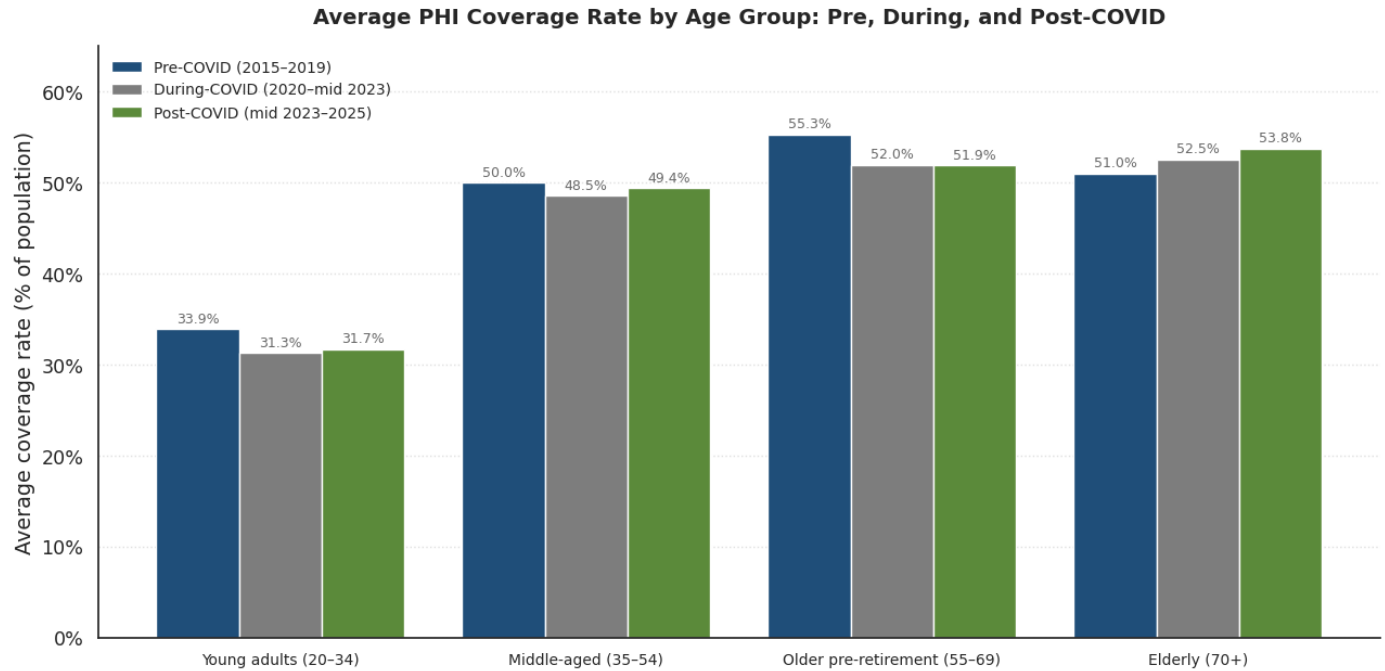


Table 2: Average PHI Coverage Rate by Age Group and Period

period	Pre-COVID (2015–2019)	During-COVID (2020–mid 2023)	Post-COVID (mid 2023–2025)	Change Pre → During (pp)	Change During → Post (pp)	Net Change Pre → Post (pp)
age_band						
Young adults (20–34)	33.88%	31.26%	31.67%	-2.62 pp	+0.41 pp	-2.21 pp
Middle-aged (35–54)	49.96%	48.54%	49.38%	-1.42 pp	+0.84 pp	-0.58 pp
Older pre-retirement (55–69)	55.27%	51.96%	51.89%	-3.31 pp	-0.07 pp	-3.38 pp
Elderly (70+)	50.98%	52.52%	53.76%	+1.54 pp	+1.24 pp	+2.78 pp

Figure 4 and Table 2 break down the average PHI coverage rate by age group across three discrete periods bounded by the WHO's official COVID-19 emergency window (March 2020 to May 2023).

Before COVID (2015–2019), coverage levels varied significantly across age groups, older pre-retirement Australians (55–69) were the most insured at 55.27%, followed by the elderly (70+) at 50.98%, middle-aged (35–54) at 49.96%, and young adults (20–34) at the bottom with just 33.88%.

When COVID hit (2020–mid 2023), coverage fell across all groups except the elderly. The older pre-retirement group took the biggest hit, losing 3.31 percentage points, while young adults dropped 2.62 percentage points to 31.26%. The middle-aged group held up relatively well, declining by just 1.42 percentage points.

During the post-COVID period (middle of 2023 – 2025), all categories showed improvement, although not by much, in order to be able to achieve pre-pandemic numbers. Young adults managed to improve only by 0.41 percentage points, having ended up at 31.67%, far from where they began. Pre-retirees barely improved, increasing by just 0.07 percentage points, the only group to break away from this pattern and keep growing throughout the whole time frame is the elderly, whose number went up to 53.76%.

In terms of net change, the final number tells us the main story, with all categories ending up with worse results compared to the pre-COVID state, except for the elderly, who were able to increase by 2.78 percentage points.

Section 3: Discussion and Sustainability Outlook

The overall picture of the PHI trend is one of fragile stability. The market looks resilient, membership is at a record high, and the national coverage rate has nearly returned to its post-2000 peak, but beneath that headline figure, three structural concerns remain unresolved.

First, young adult coverage has not recovered to pre-COVID levels and remains on a long-term downward trend, as this group pays more in premiums than it claims; their continued exit weakens the financial foundation of the entire system. Building on this, the insured pool is steadily ageing, with the elderly now the most insured age group for the first time in the dataset. This shift is consistent with the death spiral mechanism, where higher average claims push premiums up,

driving further exits among younger members (Duckett & Moran, 2021). Finally, government policy incentives have weakened in real terms, reducing the financial motivation to insure for many households (Bilgrami et al., 2021).

Without targeted policy action to improve affordability and strengthen the value proposition for younger Australians, the PHI market risks entering a slow but sustained decline, placing growing pressure on the public healthcare system over the long term.

Task 2: Statistical model to analyse PHI uptake and assess the influence of relevant factors.

This section uses individual-level survey data collected from a nationally representative sample of 1,705 Australians in 2025 to identify the factors that influence the decision to hold private hospital insurance. The analysis then uses the LPM to generate predictions, assess its limitations, and discuss how the Medicare Levy Surcharge affects demand for PHI.

Section 2.1: Regression Model, Determinants of PHI Uptake.

This section estimates the linear regression model for explaining PHI uptake at the individual level. The dependent variable is PHI, which is binary (1 if the individual holds PHI, 0 otherwise), so the model (LPM) will estimate coefficients that are interpreted as changes in the probability of holding PHI.

Linear Probability Model (LPM)

Equation of the population regression model:

$$PHI_i = \beta_0 + \beta_1 HHincome_i + \beta_2 age_i + \beta_3 age_i^2 + \beta_4 bachelor_i + \beta_5 employed_i + \beta_6 health_i + \beta_7 couple_i + \beta_8 depchildren_i + \beta_9 female_i + \beta_{10} city_i + u_i$$

The 10 explanatory variables

Order	Variable	Type	Expected sign
1	HHincome	Continuous	(+) Richer → more PHI
2	age	Continuous	(+) Older → more PHI
3	age ²	Continuous (squared)	(-) Captures the curvature (turning point)
4	employed	Binary	(+) Stable income
5	bachelor	Binary	(+) Education effect
6	depchildren	Discrete (1–4)	(+) Family responsibilities
7	couple	Binary	(+) Joint policies
8	female	Binary	(?) Likely insignificant
9	health	Ordinal (1=best, 5=worst)	(-) Better health (low number) → more PHI
10	city	Binary	(+) Better hospital access

Linear Probability Model Result Table

Table 1 — Linear Probability Model: Determinants of PHI Uptake (n = 1,705; HC1 robust standard errors)

Variable	Coefficient	Std. Error	t-statistic	p-value	95% CI Lower	95% CI Upper	Sig.
Intercept (β_0)	-0.0763	0.1099	-0.694	0.4876	-0.2917	+0.1391	
HHincome	+0.0022	0.0002	+11.071	0.0000	+0.0018	+0.0026	***
age	+0.0074	0.0041	+1.796	0.0726	-0.0007	+0.0155	*
age_sq	-0.0000	0.0000	-0.006	0.9950	-0.0001	+0.0001	
employed	+0.0679	0.0282	+2.407	0.0161	+0.0126	+0.1232	**
bachelor	+0.1772	0.0253	+7.012	0.0000	+0.1276	+0.2267	***
depchildren	-0.0310	0.0122	-2.547	0.0109	-0.0549	-0.0071	**
couple	+0.0472	0.0251	+1.877	0.0605	-0.0021	+0.0964	*
female	+0.0407	0.0220	+1.852	0.0641	-0.0024	+0.0838	*
health	-0.0647	0.0117	-5.537	0.0000	-0.0876	-0.0418	***
city	+0.0153	0.0229	+0.666	0.5053	-0.0297	+0.0602	

Table 2 — Model Fit Statistics

Statistic	Value
Observations	1,705
R-squared	0.2087
Adjusted R-squared	0.2040
F-statistic	64.91
F-statistic p-value	6.24e-112
Standard error type	HC1 (heteroskedasticity-robust)

Summary of reasoning for variable selection and interpretation of the estimated coefficients, for all explanatory variables included in our model:

Table 1 : *HHincome*

Questions	Answers
Variable	HHincome (Household income)
How variable is constructed ?	Used as provided in the dataset. No transformation applied. Continuous variable measured in thousands of Australian dollars (range \$0–\$374k; mean $\approx$ \$121k).
Why included (rationale) ?	Household income is the single most consistently identified predictor of PHI uptake (Palangkaraya & Yong, 2005; Bilgrami et al., 2021). PHI is a discretionary, premium-based product, so affordability fundamentally shapes who can purchase it. Income also activates the Medicare Levy Surcharge (MLS), which imposes a tax penalty on higher earners without PHI, creating a strong financial incentive concentrated above the income thresholds.
Expected direction (& reasoning)	Positive (+). Higher-income households can more easily afford PHI premiums, and the MLS creates a strong financial incentive for high earners to take out cover to avoid the surcharge.
Actual direction (& explanation)	Positive (+). The estimated relationship matches expectations: as income rises, the probability of holding PHI increases monotonically. This is consistent with both the affordability mechanism and the income-based design of the MLS.
Magnitude of coefficient	$\beta = +0.0022$. A \$1,000 increase in household income is associated with a 0.22 percentage point rise in the probability of holding PHI, holding all other variables constant. Equivalently, a \$50,000 increase in household income raises uptake probability by approximately 11 percentage points, a substantial effect that confirms the central role of affordability.
Sig. at 10%?	Yes, $p < 0.001$ (significant at the 1% level, ***)

Table 2 : *Age*

Questions	Answers
Variable	Age (years)
How variable is constructed	Used as provided in the dataset. Continuous variable (range 24–99 years; mean ≈ 53).
Why included (rationale)	Age is a robust predictor of PHI uptake because older individuals expect to use hospital services more intensively and therefore derive greater value from cover (Johar et al., 2011). Age also enters the policy framework through the Lifetime Health Cover (LHC) loading, which penalises individuals who delay purchasing PHI beyond age 31.
Expected direction (& reasoning)	Positive (+). Older people use hospital services more, face stronger financial incentives through the LHC loading, and are closer to age-related health events, all of which increase the expected value of holding PHI.
Actual direction (& explanation)	Positive (+). The estimated relationship matches expectations: each additional year of age is associated with a higher probability of holding PHI.
Magnitude of coefficient	$\beta = +0.0074$. Each additional year of age raises the probability of holding PHI by 0.74 percentage points, holding all else constant. This estimate must be read jointly with the age ² coefficient (Row 3) to interpret the overall age–PHI relationship.
Sig. at 10%?	Yes, $p = 0.073$ (significant at the 10% level,*)

Table 3: Age^2 (*age_sq*)

Questions	Answers
Variable	Age_sq (age squared)
How variable is constructed	Constructed as a new variable equal to the square of age. Created in Python as <code>phi_ind['age_sq'] = phi_ind['age'] ** 2</code> . Resulting values range from 576 (age 24) to 9,801 (age 99).
Why included (rationale)	Age was specified in quadratic form ($age + age^2$) to allow the model to capture potential non-linearity in the relationship between age and PHI uptake. This functional form reflects standard practice in applied econometric studies of health insurance demand. The statistical significance of the age^2 coefficient will indicate whether non-linearity is empirically supported in this dataset.
Expected direction (& reasoning)	Negative (–) , if uptake follows an inverted-U pattern in which PHI uptake rises with age but slows or reverses at the oldest ages (e.g., very elderly individuals on age pensions may downgrade cover).
Actual direction (& explanation)	Negative (–), but effectively zero. The estimated coefficient is so small that the age–PHI relationship is essentially linear over the observed age range. The data do not statistically support curvature.
Magnitude of coefficient	$\beta \approx -2.4 \times 10^{-7}$. The estimated curvature is negligible. While the aggregate APRA data in Task 1 suggested non-linear patterns across age bands, the individual-level regression provides no statistical evidence of curvature once other characteristics are controlled for. This is itself a finding worth reporting: testing for non-linearity is sound methodology, even when the test rejects the existence of curvature.
Sig. at 10%?	No , $p = 0.995$

Table 4: *Employed*

Questions	Answers
Variable	Employed (Employment status, binary: 1 = employed, 0 = not employed)
How variable is constructed	Used as provided in the dataset. Binary indicator with no transformation applied. In the sample, 62% of individuals are employed.
Why included (rationale)	Employment is a standard predictor of PHI (Palangkaraya & Yong, 2005; Buchmueller et al., 2021). Employment matters for two distinct reasons: it provides the stable household income needed to sustain ongoing premium payments, and employed individuals are more likely to be exposed to the Medicare Levy Surcharge once they cross income thresholds. Employment also captures unobserved characteristics such as financial stability and long-term planning.
Expected direction (& reasoning)	Positive (+) . Employed individuals have a more stable income stream to fund PHI premiums and face stronger incentives through the MLS. They are also more likely to have access to PHI through workplace or industry-based group policies.
Actual direction (& explanation)	Positive (+) . The estimated relationship matches expectations: employed individuals are more likely to hold PHI than those not employed, even after controlling for household income. This suggests employment captures something <i>beyond</i> income alone, likely financial stability and forward-looking behaviour.
Magnitude of coefficient	$\beta = +0.0679$. Being employed is associated with a 6.79 percentage point higher probability of holding PHI, compared with someone not employed, holding all other variables constant. This is a meaningful effect on top of the income channel.
Sig. at 10%?	Yes , $p = 0.016$ (significant at the 5% level, **)

Table 5: *Bachelor*

Questions	Answers
Variable	Bachelor (Bachelor's degree or higher, binary: 1 = yes, 0 = no)
How variable is constructed	Used as provided in the dataset. Binary indicator with no transformation applied. In the sample, 30% of individuals hold a bachelor's degree or higher.
Why included (rationale)	Education is one of the most robust predictors of PHI uptake identified in every Australian study reviewed (Hopkins & Kidd, 1996; Bilgrami et al., 2021; Kettlewell & Zhang, 2024). It captures three distinct effects: it proxies for unobserved permanent income that goes beyond what current household income can capture; it reflects greater health literacy and risk awareness ; and it is associated with stronger long-term financial planning behaviour.
Expected direction (& reasoning)	Positive (+) . Higher-educated individuals tend to have higher lifetime earnings, better understanding of insurance products, and greater awareness of the financial risks of being uninsured for major medical events.
Actual direction (& explanation)	Positive (+) . The estimated relationship matches expectations and is one of the strongest effects in the model. Bachelor-degree holders are substantially more likely to hold PHI than those without, even after controlling for income and employment.
Magnitude of coefficient	$\beta = +0.1772$. Holding a bachelor's degree raises the probability of having PHI by 17.72 percentage points, compared with someone without one, holding all other variables constant. This is the largest single binary effect in the model and confirms education as a major non-financial driver of PHI uptake.
Sig. at 10%?	Yes — $p < 0.001$ (significant at the 1% level, ***)

Table 6 : *Depchildren*

Questions	Answers
Variable	Depchildren (Number of dependent children in household)
How variable is constructed	Used as provided in the dataset. Discrete integer variable ranging from 1 to 4. Note: The minimum value is 1 rather than 0, meaning the dataset does not contain individuals with zero dependent children. The variable is interpreted accordingly, as a measure of how many dependents are in households that already have at least one, not as a presence/absence indicator.
Why included (rationale)	Family composition is a long-standing determinant of PHI demand (Palangkaraya & Yong, 2005). Parents may value PHI to ensure access to private hospitals, choice of paediatrician, and shorter waiting times for their children. Dependent children also affect household budget constraints and risk preferences.
Expected direction (& reasoning)	Positive (+) . Standard theory predicts that more dependents increase the value of PHI because families face higher expected healthcare needs and value certainty of access.
Actual direction (& explanation)	Negative (–) — opposite to the expected sign. Each additional dependent child is associated with a <i>lower</i> probability of holding PHI. This counter-intuitive result likely reflects a budget-constraint effect : larger families face higher household costs (food, schooling, clothing) which compete with PHI premiums for limited disposable income. The negative sign may also partly reflect the dataset's restriction to households with at least one dependent, which limits the interpretation.
Magnitude of coefficient	$\beta = -0.0310$. Each additional dependent child is associated with a 3.10 percentage point lower probability of holding PHI, holding all other variables (including income) constant.
Sig. at 10%?	Yes , $p = 0.011$ (significant at the 5% level, **)

Row 7: *Couple*

Questions	Answers
Variable	Couple (Couple status, binary: 1 = in a couple, 0 = not in a couple)
How variable is constructed	Used as provided in the dataset. Binary indicator with no transformation applied. In the sample, 66% of individuals are in a couple.
Why included (rationale)	Family structure is a long-standing predictor of PHI demand, as Palangkaraya and Yong (2005) and Bilgrami et al. (2021) stated in both document, married and partnered individuals are systematically more likely to hold PHI than single individuals, even after controlling for income and age. Couples often share a single joint PHI policy, which lowers the effective per-person cost, and partnered households tend to engage in more long-term financial planning, including risk protection through insurance.
Expected direction (& reasoning)	Positive (+) . Couples benefit from joint policies (lower per-person premium), pool risks across two adults, and are more likely to plan for future healthcare needs together.
Actual direction (& explanation)	Positive (+) . The estimated relationship matches expectations: individuals in a couple are more likely to hold PHI than single individuals, controlling for other characteristics.
Magnitude of coefficient	$\beta = +0.0472$. Being in a couple is associated with a 4.72 percentage point higher probability of holding PHI compared with single individuals, holding all other variables constant.
Sig. at 10%?	Yes — $p = 0.061$ (significant at the 10% level, *)

Table 8: *Female*

Questions	Answers
Variable	Female (Gender, binary: 1 = female, 0 = male)
How variable is constructed	Used as provided in the dataset. Binary indicator with no transformation applied. In the sample, 58% of individuals are female.
Why included (rationale)	Gender is included as a demographic control rather than as a variable of strong theoretical interest. Empirical PHI studies typically include it because women have, on average, higher healthcare utilisation across the lifecycle (particularly in reproductive years) and may therefore place higher value on PHI cover. The literature is mixed on the size and significance of the effect in Australia (Hopkins & Kidd, 1996), but omitting gender would risk attributing female-specific effects to other variables.
Expected direction (& reasoning)	Positive (+), but uncertain. Higher female healthcare utilisation suggests a positive sign, though the effect is often small once age and income are controlled for.
Actual direction (& explanation)	Positive (+). The estimated effect is positive but small, consistent with the mixed evidence in the literature. Women in the sample are slightly more likely to hold PHI than men, all else equal.
Magnitude of coefficient	$\beta = +0.0407$. Being female is associated with a 4.07 percentage point higher probability of holding PHI compared with being male, holding all other variables constant.
Sig. at 10%?	Yes — $p = 0.064$ (significant at the 10% level, *)

Table 9: *Health (the most interesting variable in the model)*

Questions	Answers
Variable	Health (Self-reported health status, ordinal: 1 = excellent, 5 = poor)
How variable is constructed	Used as provided in the dataset. Ordinal variable on a 1–5 scale where lower values indicate better health .
Why included (rationale)	Health status is central to the economic theory of insurance demand, particularly adverse selection theory (Barrett & Conlon, 2003). The classical prediction is that individuals with worse health value insurance more highly because they expect higher medical costs. Health status is therefore a critical variable for testing whether adverse selection operates in the Australian PHI market.
Expected direction (& reasoning)	Negative (–) . Under classical adverse selection, sicker individuals (higher health values) would be <i>more</i> likely to buy PHI, producing a positive coefficient on health. However, in the Australian context, the opposite is widely expected: the Medicare Levy Surcharge forces high-income (typically healthier) individuals into PHI, and community rating prevents insurers from price-discriminating against the sick.
Actual direction (& explanation)	Negative (–) . The estimated relationship is strongly negative and matches the Australian-context expectation. Better health (lower score) is associated with substantially higher PHI uptake. This empirically supports the view that policy design has overridden the textbook adverse-selection mechanism in the Australian market — the <i>healthy</i> are concentrated in PHI, not the sick.
Magnitude of coefficient	$\beta = -0.0647$. A one-unit deterioration in self-reported health (e.g. moving from "excellent" to "very good") is associated with a 6.47 percentage point <i>decrease</i> in the probability of holding PHI, holding all other variables constant. Across the full scale (from 1 to 5), this translates to a difference of roughly 26 percentage points in PHI probability between the healthiest and least healthy individuals — one of the largest effects in the model.
Sig. at 10%?	Yes — $p < 0.001$ (significant at the 1% level, ***)

Table 10: *City*

Column	Content
Variable	City (Urban residence, binary: 1 = lives in a major city, 0 = lives elsewhere)
How variable is constructed	Used as provided in the dataset. Binary indicator with no transformation applied. In the sample, 65% of individuals live in a major city.
Why included (rationale)	Geographic location is included to control for unequal access to the private hospital system across Australia. Private hospital infrastructure is heavily concentrated in metropolitan areas, while regional and remote populations rely more on the public hospital network. As a result, the <i>practical value</i> of holding PHI may be higher for urban residents who have greater access to private facilities and specialists. While not a central variable in the Australian PHI literature, location is a standard demographic control that captures supply-side variation in access.
Expected direction (& reasoning)	Positive (+) . Urban residents have better physical access to private hospitals and a wider choice of specialists, increasing the perceived value of holding PHI.
Actual direction (& explanation)	Positive (+) , but small. The coefficient has the expected sign, but the effect is not statistically distinguishable from zero. This suggests that, once income, education and other characteristics are controlled for, geographic location adds little additional predictive power to the model.
Magnitude of coefficient	$\beta = +0.0153$. Living in a major city is associated with a 1.53 percentage point higher probability of holding PHI compared with living elsewhere, holding all other variables constant — a small effect by the standards of the model.
Sig. at 10%?	No — $p = 0.505$

Section 2.2 : Missing Variables

While the PHI_2026 dataset contains a strong set of personal and household characteristics, several determinants of PHI uptake are not directly observed, four omissions stand out.

First, the dataset contains no information on the price of PHI faced by each individual, premium level, level of cover (basic, mid, top), or chosen excess. Price is one of the strongest determinants of insurance demand (Kettlewell & Zhang, 2024; Bilgrami et al., 2021), and its absence prevents the model from separating willingness to pay from ability to pay.

Second, there is no measure of Medicare Levy Surcharge (MLS) liability, the MLS depends on individual taxable income and family thresholds rather than on household income, yet it is a key driver of PHI uptake at the top of the income distribution (Buchmueller et al., 2021). Its absence means the income coefficient mixes affordability and tax incentive effects.

Third, the dataset omits measures of family medical history, expected healthcare needs, and risk aversion. These are theoretically central to insurance demand and their absence likely contributes to omitted variable bias on the health coefficient.

Finally, the dataset does not capture employer-provided PHI benefits or workplace group schemes, which lower the effective price of cover for many Australian employees and shift the take-up decision.

Section 2.3 Omitted Variable Bias and Reverse Causality

The estimated coefficients describe statistical associations rather than causal effects, since causal interpretation requires both no omitted determinants of PHI uptake and one-way causality from the explanatory variables to PHI, three variables are particularly vulnerable.

Health is affected by both problems. The model does not capture risk aversion, family medical history, or expected healthcare needs, traits that may produce both worse self-reported health and stronger demand for PHI, biasing the coefficient. Reverse causality also applies, holding PHI itself can improve reported health through faster specialist access and preventive screening (Johar et al., 2011).

Bachelor's degree is primarily exposed to omitted variable bias. Education correlates with unobserved traits such as financial literacy, long-term planning, family background, and risk aversion, which independently influence PHI uptake. Part of the estimated +17.7 pp effect therefore likely reflects these traits rather than education itself.

Employment is exposed to both problems. It correlates with unobserved health (sicker individuals are less likely to work) and with workplace PHI subsidies not captured in the dataset. It is also subject to mild reverse causality, the financial commitment of holding PHI may motivate individuals, particularly at older ages, to remain employed.

The estimated coefficients should therefore be interpreted as associations rather than causal effects. Establishing causality would require richer data capturing risk aversion, family history, and workplace benefits.

Section 2.4: Creation of synthetic individual with average characteristics of the variables included your LPM.

Using the estimated Linear Probability Model, this section predicts the probability of holding PHI for a synthetic average individual, a hypothetical person whose characteristics equal the sample mean of each explanatory variable. The probability is then recomputed under three counterfactual scenarios.

Build the synthetic average individual:

Table 1 - Synthetic Average Individual: Mean Values of Each Variable	
Variable	Average value
Household income (\$000s)	120.94
Age (years)	53.09
Age squared	3072.05
Employed (share)	0.62
Bachelor's degree (share)	0.30
Dependent children (avg.)	1.60
In a couple (share)	0.66
Female (share)	0.58
Self-reported health (1=best, 5=worst)	2.69
Lives in major city (share)	0.65

Description of the synthetic individual

The average individual in the sample is approximately 53 years old, with a household income of \$121,000, in moderately good self-reported health (2.69 on the 1–5 scale, where 1 is best), and with 1.6 dependent children on average. They are 62% likely to be employed, 66% likely to be in a couple, 58% likely to be female, 65% likely to live in a major city, and 30% likely to hold a bachelor's degree.

Table 2 - Baseline Predicted Probability for the Synthetic Average Individual

Metric	Value
Predicted probability of holding PHI	0.5182
Equivalent percentage	51.8%

Plugging these values into the estimated regression equation yields a predicted probability of holding PHI of **0.518 (51.8%)**. As expected, this baseline figure is very close to the overall PHI uptake rate observed in the sample (52%), which confirms that the model is internally consistent at the means.

Results

Table 3 — Predicted PHI Probability under Counterfactual Scenarios

Scenario	What changes	P(PHI)	P(PHI), %	Δ vs baseline (pp)
Baseline (average individual)	—	0.5182	51.8%	+0.00
A) Household income doubles	HHincome: \$121k → \$242k	0.7845	78.4%	+26.63
B) 10 years older	age: 53 → 63 (age ² updated)	0.5920	59.2%	+7.38
C) Obtains a bachelor's degree	bachelor: 0.30 → 1	0.6422	64.2%	+12.40

Doubling household income from \$121k to \$242k raises the predicted probability of PHI by 26.6 percentage points, from 51.8% to 78.4%, by far the largest effect, confirming income as the dominant driver of PHI uptake through both affordability and MLS exposure. Becoming ten years

older (53 to 63) raises the probability by 7.4 percentage points (51.8% to 59.2%), a meaningful but smaller effect, consistent with the modest age coefficient and the absence of significant curvature. Obtaining a bachelor's degree raises the probability by 12.4 percentage points (51.8% to 64.2%), the second-largest effect, confirming education as a major non-financial determinant of PHI uptake.

Together, these scenarios show that the strongest predicted changes in PHI uptake come from income and education, with age playing a more limited role for an already middle-aged individual.

Section 2.5 Estimating the Effect of the Medicare Levy Surcharge

The Medicare Levy Surcharge (MLS) is a tax penalty applied to higher-income individuals without PHI, activated at specific income thresholds. Estimating its causal effect requires a design that separates the tax-incentive effect from other factors varying with income.

A regression-discontinuity (RDD) approach is the most natural strategy. Comparing PHI uptake just above and just below the threshold isolates the MLS effect, since individuals on either side are similar in underlying characteristics but differ sharply in MLS exposure. Stavrunova and Yerokhin (2014) apply an RDD to 2008 Australian tax return data and find that PHI coverage rises by approximately 6.5 percentage points at the threshold, with the effect strongest among young taxpayers.

An alternative is a panel-data approach exploiting within-individual variation over time. Buchmueller et al. (2021) use the HILDA longitudinal survey, leveraging year-to-year income fluctuations and a 2008 reform that raised the MLS thresholds, a design closer in to difference in differences.

Three assumptions must hold, for a valid causal interpretation. First, there must be no precise manipulation of income around the threshold; Stavrunova and Yerokhin (2014) themselves document "bunching" of reported income just below the cutoff, complicating identification. Second, all other determinants of PHI uptake must vary smoothly across the threshold (for RDD) or be appropriately controlled for (for the panel-data approach). Third, the estimated effect must

be isolated from concurrent policy changes (such as the PHI Rebate or LHC) that also influence uptake.

If these assumptions hold, the estimate captures the causal effect of the MLS on PHI demand. If they fail, the estimate mixes MLS exposure with underlying income-related preferences, and should be interpreted as an association rather than a causal impact.

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